

EXP 7: HOUGH-TRANSFORM

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In [1]: import cv2
import numpy as np
import matplotlib.pyplot as plt

In [2]: # Step 2: Load the image using imread() from cv2 module
image = cv2.imread('pic .jpg') # Replace 'image.jpg' with your image path

In [3]: # Step 3: Convert the image to grayscale
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

In [4]: # Input image and grayscale image
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB)) # Convert image to RGB for disp
plt.title("Input Image")
plt.axis('off')
```

Out[4]: (np.float64(-0.5), np.float64(735.5), np.float64(367.5), np.float64(-0.5))

Input Image



```
In [5]: plt.imshow(gray_image, cmap='gray')
plt.title("Grayscale Image")
plt.axis('off')
```

Out[5]: (np.float64(-0.5), np.float64(735.5), np.float64(367.5), np.float64(-0.5))

Grayscale Image

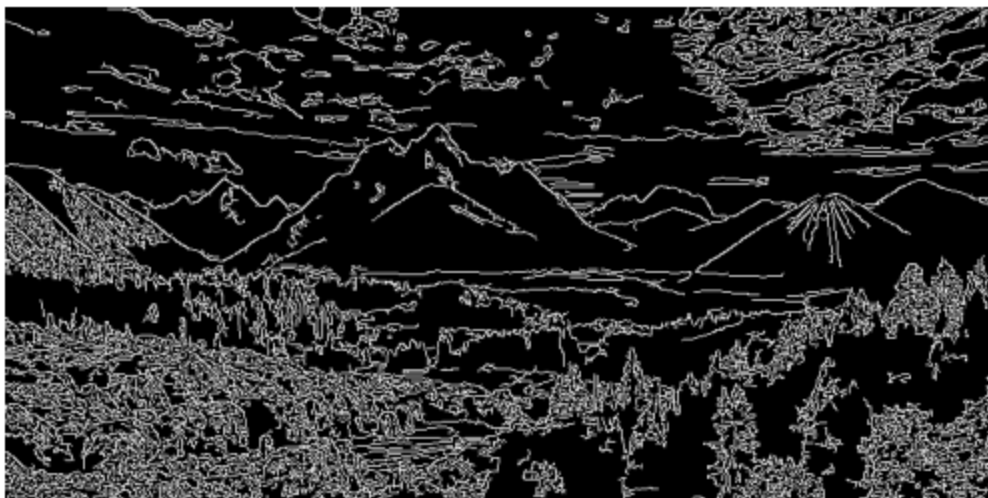


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In [6]: # Step 4: Using Canny operator from cv2, detect the edges of the image
edges = cv2.Canny(gray_image, 50, 150) # Canny edge detection with threshold value
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In [7]: # Canny Edge Detector output
plt.imshow(edges, cmap='gray')
plt.title("Canny Edge Detector")
plt.axis('off')
```

```
Out[7]: (np.float64(-0.5), np.float64(735.5), np.float64(367.5), np.float64(-0.5))
```

Canny Edge Detector



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In [8]: # Step 5: Using the HoughLinesP(), detect line coordinates for every point in the i
# The parameters of HoughLinesP are: image, resolution, threshold, minLineLength, m
lines = cv2.HoughLinesP(edges, 1, np.pi / 180, 100, minLineLength=50, maxLineGap=10)
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In [9]: # Step 6: Using a for loop, draw the lines on the original image using the detected
# The lines variable contains the endpoints of the detected lines
for line in lines:
    x1, y1, x2, y2 = line[0] # Unpacking the line coordinates
    cv2.line(image, (x1, y1), (x2, y2), (0, 255, 0), 2) # Draw green lines with th
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In [10]: # Display the result of Hough Transform (Image with lines)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB)) # Image with lines drawn
plt.title("Result of Hough Transform")
plt.axis('off')
```

```
Out[10]: (np.float64(-0.5), np.float64(735.5), np.float64(367.5), np.float64(-0.5))
```

Result of Hough Transform

